IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

ANALYSIS AND DESIGN OF CIRCUITS

Wednesday, 28 April 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions. Question One carries 40% of the marks. Question 2 and 3 carry equal marks (30% each)

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A.S. Holmes, P.D. Mitcheson
Second Marker(s) : P. Georgiou, S. Lucyszyn

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Information for Candidates

Numerical answers

- Numerical answers should be given as fully evaluated decimal values and not as unevaluated arithmetic expressions.

Notation

The following notation is used in this paper:

- The voltage waveform at node X in an AC circuit is denoted by $x(t)$ while the phasor voltage is denoted by X.

- Component and source values are generally given in Ohms, Farads, Henrys, Volts or Amps with the unit symbol omitted. Where an imaginary number is specified, it represents the complex impedance or phasor value.

- Times are in seconds unless otherwise stated.

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The Questions

1. a) Using nodal analysis, calculate the voltages at nodes *X* and *Y* of Figure 1.1. [6]

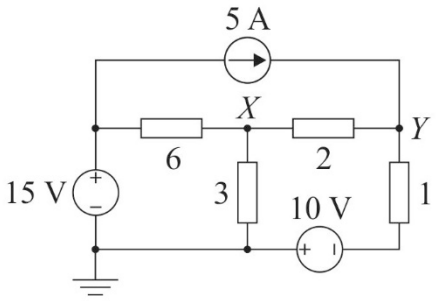


Figure 1.1

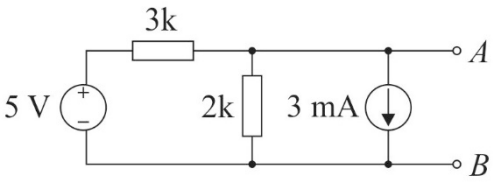


Figure 1.2

b) Draw the Thévenin equivalent of the network shown in Figure 1.2, taking care to show clearly the sign of the voltage source. [5]

c) For the circuit in Figure 1.3, write down an expression relating *Z* to *X* and *Y*. You should assume the op-amp is ideal. [5]

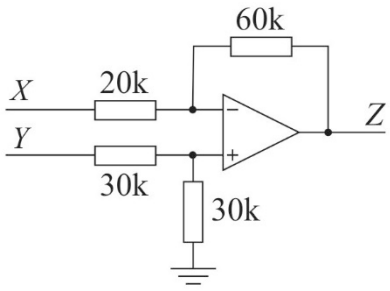


Figure 1.3

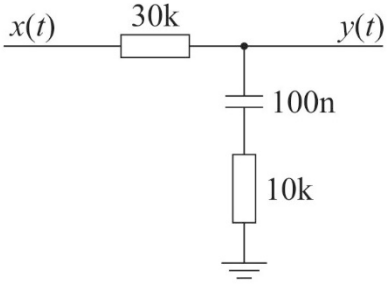


Figure 1.4

d) Write down the transfer function $\frac{Y}{X}(j\omega)$ for the circuit in Figure 1.4. Using the transfer function, or otherwise, derive an expression for the response $y(t)$ for $t > 0$ when $x(t)$ has the form:

$$x(t) = \begin{cases} 0 & t < 0 \\ 2 & t \geq 0 \end{cases}.$$ [8]

Question 1 continues on the next page…

Question 1 continued

e) For the circuit in Figure 1.5, choose the values of R_E and R_C to give a collector bias current of 0.4 mA and a collector voltage of 6 V. State clearly any assumptions made in your calculations. [5]

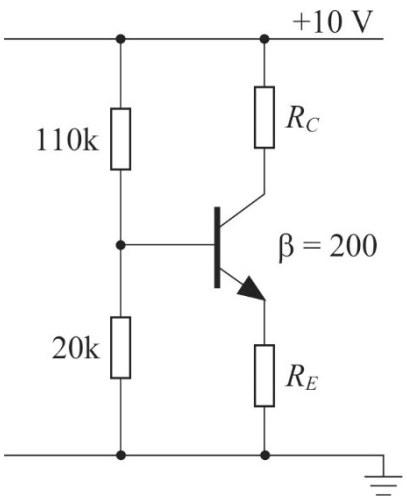


Figure 1.5

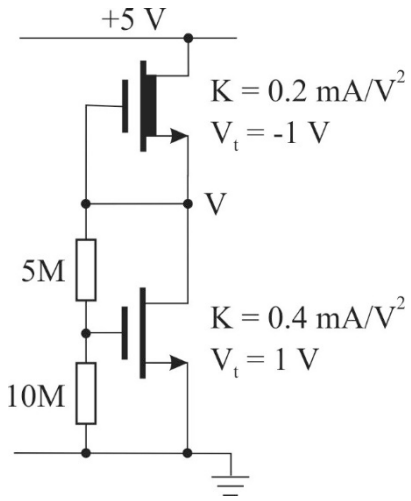


Figure 1.6

f) Calculate the voltage V in the circuit of Figure 1.6, assuming the MOSFETs have infinite output resistance. State any other assumptions made in your calculations. [5]

g) Using the resistance reflection rule, or otherwise, determine the small-signal output resistance of the circuit in Figure 1.7. [6]

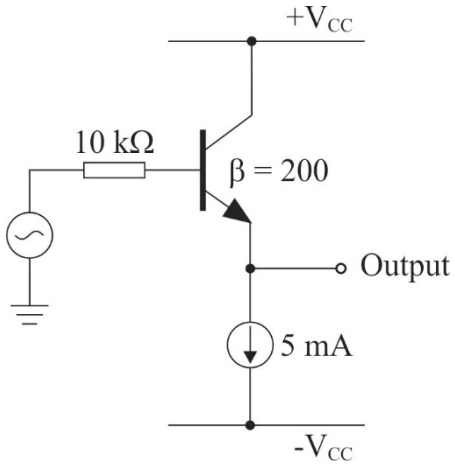


Figure 1.7

2. In the circuits shown below, U_1 and U_2 are single-supply-rail op-amps with rail-to-rail input and output voltage ranges. You may assume the op-amps are ideal apart from having limited output voltage swing and input voltage range.

a) For the circuit in Figure 2.1, show that the voltages at *X* and *Y* satisfy the relation:

$$\frac{dy}{dt} = \frac{1}{2}\frac{dx}{dt} - \frac{x}{2R_1C}$$

[8]

b) For the comparator circuit in Figure 2.2, show that the output voltage at *Z* is given by:

$$z = \begin{cases} V_{DD} & y < (1-k)V_{DD} \\ 0 \text{ or } V_{DD} & (1-k)V_{DD} < y < k\,V_{DD} \\ 0 & y > kV_{DD} \end{cases},$$

where V_{DD} is the supply voltage, and derive an expression for *k* in terms of R_A and R_C. What determines the value taken by *z* when $(1-k)V_{DD} < y < kV_{DD}$? [8]

c) The circuits in Figures 2.1 and 2.2 are combined with a MOSFET and one additional resistor to form the circuit shown in Figure 2.3. This circuit is a voltage-controlled oscillator in which the oscillation frequency is proportional to the DC control voltage applied at *X*. You may assume the MOSFET channel has infinite resistance when *z* = 0 and zero resistance when *z* = 5 V.

Explain how the oscillator works, including in your answer sketches showing the waveforms you would expect at *Y* and *Z* when the control voltage is constant. [8]

d) Obtain an expression for the oscillation frequency, and hence choose the value of *C* such that the frequency is 2.5 kHz when *x* = 2.5 V. [6]

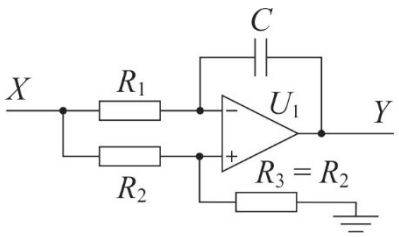


Figure 2.1

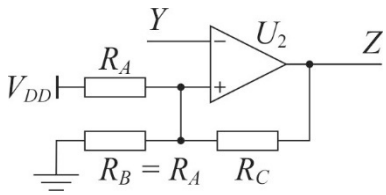


Figure 2.2

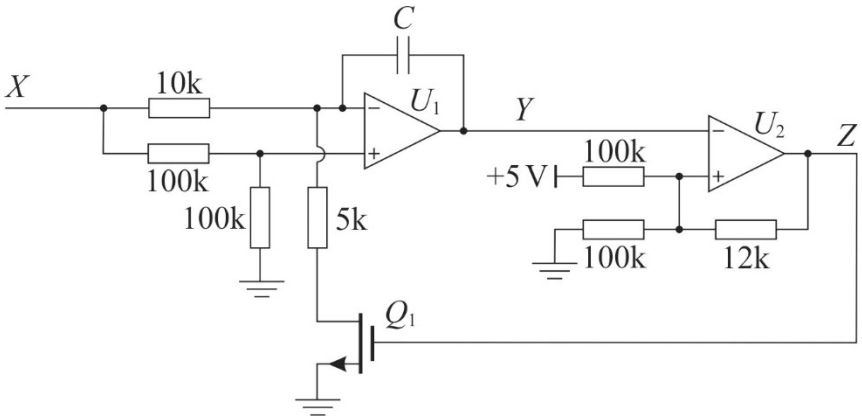


Figure 2.3

3. a) Explain (without drawing a diagram) how the circuit of Figure. 3.1 achieves a low common mode gain from the inputs v_{in1} and v_{in2} to the single output terminal, v_{out}. [4]

b) Choose a common value for R_1 and R_2 so that the quiescent value of v_{out} is at 5 V. [2]

c) Draw a suitable small signal equivalent circuit of Figure 3.1 and calculate the single-ended differential-mode gain, the common-mode gain, and the common mode rejection ratio. Assume all the transistors are identical and state any assumptions you make. Assume $v_d = (v_{in1} - v_{in2})$. [7]

d) What is the common-mode voltage range of the circuit in Figure 3.1 and why is it inadvisable to operate the circuit outside of these limits? What happens if these limits are exceeded? [4]

e) The output of Figure 3.1 is to be capacitively coupled to the input terminal of the circuit of Figure 3.2. Calculate the voltage gain and input resistance of the circuit in Figure 3.2 in isolation. You may ignore the output resistance of the transistor in the calculation and assume that the 215 kΩ resistor is open circuit in the small signal analysis.

[6]

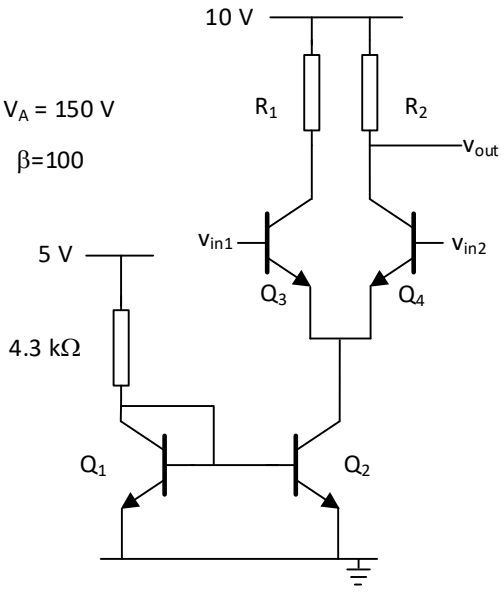


Figure 3.1

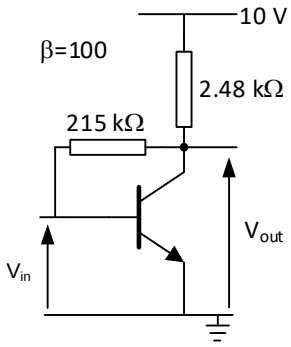


Figure 3.2

f) By drawing a macro-model, calculate the overall mid-band gain of the system when Figures 3.1 and 3.2 are capacitively coupled with a 100 nF capacitor. What is the lower corner frequency of the mid-band?

[7]